

Refined 3D Index — Full Report

5.1

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1 Manifold Overview

Manifold name: **5_1**

- Number of ideal tetrahedra: $n = 3$
- Number of cusps: $r = 1$
- Internal edges: $n - r = 2$ (easy: 1, hard: 1)
- Computation order: $q^{24/2}$
- Boundary curve range: $m \in \{-2, -1, 0, 1, 2\}$, $e \in \{-2, -\frac{3}{2}, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, 2\}$

2 Ideal Triangulation (SnaPy Data)

Gluing equation matrix from SnaPy, shape (5×9) .

Edge	(Z_1, Z'_1, Z''_1)	(Z_2, Z'_2, Z''_2)	(Z_3, Z'_3, Z''_3)
0	$(1, 0, 1)$	$(1, 2, 0)$	$(1, 1, 0)$
1	$(0, 2, 1)$	$(1, 0, 2)$	$(0, 1, 2)$
2	$(1, 0, 0)$	$(0, 0, 0)$	$(1, 0, 0)$

Cusp	(Z_1, Z'_1, Z''_1)	(Z_2, Z'_2, Z''_2)	(Z_3, Z'_3, Z''_3)
α	$(0, 0, -1)$	$(0, -1, 0)$	$(0, 0, 1)$
β	$(1, 1, -7)$	$(0, -10, 0)$	$(-1, 1, 9)$

3 Edge Classification

Easy edges: 1 total, 1 independent. Hard edges: 1.

Hard Edges			
Index	Symbolic Equation	Triplets (f_i, g_i, h_i)	
H_0	$2Z'_1 + Z''_1 + Z_2 + 2Z''_2 + Z'_3 + 2Z''_3 = 2$	$(0, 2, 1)$	$(1, 0, 2) (0, 1, 2)$

Easy Edges			
Index	Independent?	Symbolic Equation	Triplets (f_i, g_i, h_i)
E_0	✓	$Z_1 + Z_3 = 2$	$(1, 0, 0) (0, 0, 0) (1, 0, 0)$

4 Neumann–Zagier Data

$g_{\text{NZ}} \in \text{Sp}(2n, \mathbb{Q})$, 6×6 , column ordering $(Z_1, \dots, Z_n, Z'_1, \dots, Z''_n)$.

$$g_{\text{NZ}} = \begin{pmatrix} 0 & 1 & 0 & -1 & 1 & 1 \\ -2 & 1 & -1 & -1 & 2 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 5 & -1 & -4 & 5 & 4 \\ 0 & 0 & 1 & -1 & 1 & 1 \\ -2 & 0 & 0 & -1 & 2 & 2 \end{pmatrix}$$

Row labels (right margin): $\mu_0, H_0, E_0, \lambda_0/2, \Gamma_0, \Gamma_1$.

Affine shifts: $\nu_x = (-1, 1, -2)$, $\nu_p = (-4, 0, 0)$.

5 3D Index Formula

The 3D index is computed via:

$$I(\mathbf{m}_{\text{ext}}, \mathbf{e}_{\text{ext}}) = \sum_{\mathbf{e}_{\text{int}} \in (\frac{1}{2}\mathbb{Z})^{n-r}} (-q^{1/2})^{\mathbf{m} \cdot \boldsymbol{\nu}_p - \mathbf{e} \cdot \boldsymbol{\nu}_x} \prod_{a=1}^n I_{\Delta}(m_a, e_a)$$

where $\kappa = (\mathbf{m}_{\text{ext}}, \mathbf{0}^{n-r}, \mathbf{e}_{\text{ext}}, \mathbf{e}_{\text{int}})$ and the local charges are:

$$m_a = (g_{\text{NZ}}^{-1} \kappa)_a, \quad e_a = (g_{\text{NZ}}^{-1} \kappa)_{n+a}$$

Only terms with integer (m_a, e_a) contribute.

Explicit Local Charges. Δ_1 : $m_1 = -4m_0 + e_0 + e_{\text{int},0}$, $e_1 = -2e_{\text{int},0} + e_{\text{int},1}$

Δ_2 : $m_2 = 5m_0 - e_0 - 2e_{\text{int},0}$, $e_2 = -5m_0 + e_0 + e_{\text{int},0}$

Δ_3 : $m_3 = 4m_0 - e_0 - e_{\text{int},0}$, $e_3 = m_0 - e_{\text{int},0} + e_{\text{int},1}$

The tetrahedron index $I_{\Delta}(m, e)$ is given by

$$I_t(m, e) = \sum_{k=\max(0, -e)}^{\infty} \frac{(-1)^k q^{k(k+1)/2 - (2k+e)m/2}}{\prod_{j=1}^k (1 - q^j) \prod_{j=1}^{k+e} (1 - q^j)}$$

with $I_{\Delta}(m, e) = (-q^{1/2})^m \cdot I_t(-m - e, m)$ when $m + e \geq 0$, else $I_{\Delta}(m, e) = I_t(m, e)$.

6 Refined Index Results

Series truncated at $q^{24/2}$ (i.e. $N_{\text{max}} = 12$). Number of computed sectors: 45. Non-zero sectors: 1.

$$\begin{aligned}
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (-2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (-3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (-1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (-1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (0)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (-2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (-3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (-1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (-1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (0)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (-2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (-3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (-1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (-1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (0)) &= 1 + q\eta_0^{-1} - q + q^2\eta_0^{-1} - 2q^2 + q^2\eta_0 + q^3\eta_0^{-1} - 2q^3 + q^3\eta_0 + q^4\eta_0^{-2} - 2q^4 + q^4\eta_0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (-2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (-3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (-1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (-1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (0)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (-2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (-3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (-1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (-1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (0)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (1)) &= 0
\end{aligned}$$

7 Weyl Symmetry

Define the Weyl-shifted index:

$$f(\eta_j; m, e) = \eta_j^{a_j \cdot e + b_j \cdot m} \cdot I^{\text{ref}}(m, e)$$

Weyl symmetry requires $f(m, e) = f(-m, -e)$ for each sector pair, where f denotes the Weyl-shifted index above.

Weyl symmetry $f(m, e) = f(-m, -e)$: 0/45 sectors pass **FAIL** (45 failures may be due to missing partner sectors or finite truncation at large $|m|$, $|e|$).

The adjoint $\text{su}(2)$ projection (eq. 2.59–2.61) integrates out η and cusp fugacities against the Haar measure weighted by the adjoint character; the result must equal -1 per filled cusp.

Intermediate c_e values ((q^1, η^0) coefficient of $I^{\text{ref}}(m = 0, e)$):

e	c_e
-2	0
$-\frac{3}{2}$	0
-1	0
$-\frac{1}{2}$	0
0	-1
$\frac{1}{2}$	0
1	0
$\frac{3}{2}$	0
2	0

Projected value: 0 (expected -1): **FAIL**